# Decimation of a Gaussian Army

## Moment-Matching Discrete Deconvolution and Applications in Statistics, Signal Processing, and Machine Learning


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**Abstract**

Given a target normal distribution $n = \mathcal{N}(\mu, \sigma^2)$ and a "reduced" normal $m = \mathcal{N}(\mu, \sigma^2/N)$, we seek a finite discrete probability measure $d$ supported on $N$ atoms such that the convolution $m * d = n$. Taking Fourier (characteristic-function) transforms reduces this to a deconvolution problem whose solution is itself Gaussian with mean zero and variance $\sigma^2(1 - 1/N)$. Because no *finite* discrete measure can perfectly replicate a Gaussian characteristic function on all of $\mathbb{R}$, we show that the optimal $N$-point approximation—in the sense of matching all polynomial moments up to degree $2N - 1$—is furnished by the classical Gauss–Hermite quadrature rule. We derive the construction in full generality, work out the exact closed-form solution for $N = 3$, verify the moment-matching properties, quantify the residual error in the characteristic function, and connect the result to applications in Bayesian inference, federated learning, mixture modelling, and numerical quadrature. A proof-of-concept algorithm is provided alongside PGF/Ti*k*Z visualisations of the mixture approximation.

The paper ends with "The End"


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# 1 Introduction

Gaussian distributions occupy a central role in probability theory, statistics, and machine learning. The convolution of two independent Gaussian random variables is again Gaussian—a closure property that underpins the Central Limit Theorem, Kalman filtering, and the theory of exponential families. A natural inverse question is: *given a Gaussian target, can one "factor" it into a sharper Gaussian and a simple discrete distribution?*

This paper answers that question precisely. We are given:

- $n$: a *target* normal density $\mathcal{N}(\mu, \sigma^2)$;
- $m$: a *reduced* normal density $\mathcal{N}(\mu, \sigma^2/N)$ for some positive integer $N$;
- $d$: an unknown finite discrete measure on $N$ atoms.

We wish to find $d$ such that $m * d = n$. The answer connects to Gauss–Hermite quadrature [1, 2], the moment problem [5], and numerical methods for Bayesian computation [6, 7].

**Motivation.** The problem arises naturally in several applied contexts:

**Federated / distributed learning.** $N$ agents each hold a local posterior $\mathcal{N}(\mu, \sigma^2/N)$ (as if each observed $1/N$ of the data). Combining them via a discrete "consensus measure" $d$ should recover the full-data posterior $\mathcal{N}(\mu, \sigma^2)$.

**Particle filtering.** One seeks a minimal particle representation that, when convolved with a Gaussian transition kernel, reproduces a prescribed predictive distribution.

**Quadrature design.** The weights and nodes of $d$ are exactly the Gauss–Hermite quadrature points for the target Gaussian, giving an optimal integration rule.

**Numerical analysis of PDEs.** Operator splitting methods decompose diffusion (Gaussian convolution) and transport steps; $d$ encodes the transport component.

**Paper organisation.** Section 2 states the problem formally. Section 3 solves it in the Fourier domain and identifies the obstruction to an exact finite solution. Section 4 reviews Gauss–Hermite quadrature and formulates the optimal $N$-point approximation. Section 5 works out the $N = 3$ case in full detail. Section 6 verifies the moment-matching properties. Section 7 quantifies the approximation error. Section 8 presents an algorithm. Section 10 discusses applications in AI/ML. Section 11 concludes. A glossary appears in Section 11.

# 2 Problem Setup

**Definition 2.1** (Discrete probability density)**.** A *finite discrete probability density* on $N$ atoms is a signed measure of the form

$$d(x) = \sum_{i=1}^{N} h_i\, \delta(x - a_i), \qquad h_i \geq 0, \quad \sum_{i=1}^{N} h_i = 1,$$

where $\delta$ denotes the Dirac delta, $a_i \in \mathbb{R}$ are the *nodes*, and $h_i$ are the *weights* (heights).

**Definition 2.2** (Problem statement)**.** Let $N \in \mathbb{Z}_{>0}$, $\mu \in \mathbb{R}$, $\sigma > 0$. Define

$$m(x) = \frac{\sqrt{N}}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{N(x-\mu)^2}{2\sigma^2}\Big) = \phi_{\mu,\,\sigma^2/N}(x), \qquad n(x) = \phi_{\mu,\sigma^2}(x),$$

where $\phi_{\mu,v}$ denotes the $\mathcal{N}(\mu, v)$ probability density function. Find nodes $\{a_i\}_{i=1}^N$ and weights $\{h_i\}_{i=1}^N$ such that

$$(m * d)(x) \;=\; n(x) \quad \forall x \in \mathbb{R}.$$

Note that the convolution evaluates as

$$(m * d)(x) = \sum_{i=1}^{N} h_i\, m(x - a_i) = \sum_{i=1}^{N} h_i\, \phi_{\mu + a_i,\, \sigma^2/N}(x),$$

i.e. $d$ generates a *Gaussian mixture* whose components each have variance $\sigma^2/N$.

# 3 Fourier Analysis and the Deconvolution Identity

## 3.1 Characteristic functions

The characteristic function (Fourier transform of the density) of $\mathcal{N}(\mu, v)$ is

$$\hat{\phi}_{\mu,v}(\omega) = \int_{-\infty}^{\infty} e^{i\omega x}\, \phi_{\mu,v}(x)\, dx = \exp\Big(i\mu\omega - \tfrac{v}{2}\omega^2\Big).$$

## 3.2 Deconvolution in the Fourier domain

Convolution in the space domain corresponds to pointwise multiplication of characteristic functions:

$$\widehat{m * d}(\omega) = \hat{m}(\omega) \cdot \hat{d}(\omega).$$

For the equation $m * d = n$ to hold, we therefore need

$$\hat{d}(\omega) = \frac{\hat{n}(\omega)}{\hat{m}(\omega)}.$$

**Proposition 3.1** (Ideal deconvolution kernel)**.** *Under Definition 2.2,*

$$\hat{d}(\omega) = \frac{e^{i\mu\omega - \frac{\sigma^2}{2}\omega^2}}{e^{i\mu\omega - \frac{\sigma^2}{2N}\omega^2}} = \exp\Big(-\frac{\sigma^2}{2}\Big(1 - \frac{1}{N}\Big)\omega^2\Big).$$

*This is the characteristic function of* $\mathcal{N}\big(0,\ \sigma^2(1 - 1/N)\big)$.

*Proof.* Direct substitution of $\hat{m}(\omega) = e^{i\mu\omega - \sigma^2\omega^2/(2N)}$ and $\hat{n}(\omega) = e^{i\mu\omega - \sigma^2\omega^2/2}$ into the ratio and simplification gives the stated expression, which is recognised as the characteristic function of a centred Gaussian with variance $\tau^2 := \sigma^2(1 - N^{-1})$. □

## 3.3 Impossibility of an exact finite solution

**Theorem 3.2** (No exact finite discrete solution)**.** *There is no finite discrete measure* $d = \sum_{i=1}^{N} h_i\, \delta(\cdot - a_i)$ *with* $N < \infty$ *such that* $\hat{d}(\omega) = e^{-\tau^2\omega^2/2}$ *for all* $\omega \in \mathbb{R}$.

*Proof.* The characteristic function of any finite discrete measure is an *almost-periodic* function (a finite trigonometric sum):

$$\hat{d}(\omega) = \sum_{i=1}^{N} h_i\, e^{ia_i\omega}.$$

Such a function is a polynomial in $e^{i\omega}$ up to a global phase, hence it cannot vanish on a set of positive measure and cannot be entire of order 2. By contrast, $e^{-\tau^2\omega^2/2}$ is a Gaussian in $\omega$, which is entire of order 2 and decays to zero as $|\omega| \to \infty$. Since a nontrivial finite trigonometric sum cannot decay to zero as $|\omega| \to \infty$, the two cannot be equal everywhere. □

The best we can hope for is moment-matching up to a finite order, which is precisely what Gauss–Hermite quadrature achieves.

# 4 Gauss–Hermite Quadrature

## 4.1 Hermite polynomials

**Definition 4.1** (Probabilists' Hermite polynomials)**.** The probabilists' Hermite polynomials $\{He_n\}_{n\geq 0}$ are orthogonal with respect to the standard Gaussian weight $w(t) = e^{-t^2/2}$ on $\mathbb{R}$:

$$He_n(t) = (-1)^n e^{t^2/2} \frac{d^n}{dt^n} e^{-t^2/2}, \qquad He_0 = 1,\ He_1 = t,\ He_2 = t^2 - 1,\ He_3 = t^3 - 3t, \ldots$$

They satisfy the three-term recurrence $He_{n+1}(t) = t\, He_n(t) - n\, He_{n-1}(t)$.

*Remark* 4.2. The *physicists'* Hermite polynomials $H_n(t)$ used in the quadrature tables relate by $H_n(t) = 2^{n/2} He_n(\sqrt{2}\, t)$. We use physicists' conventions in Section 5 to match standard quadrature tables.

## 4.2 Gauss–Hermite quadrature rule

The $N$-point Gauss–Hermite quadrature rule provides nodes $t_1 < t_2 < \cdots < t_N$ (roots of $H_N$) and weights $w_1^{\mathrm{GH}}, \dots, w_N^{\mathrm{GH}}$ such that

$$\int_{-\infty}^{\infty} f(t)\, e^{-t^2}\, dt \;=\; \sum_{i=1}^{N} w_i^{\mathrm{GH}}\, f(t_i) \quad \text{exactly for all } f \in \mathcal{P}_{2N-1},$$

where $\mathcal{P}_{2N-1}$ denotes the space of polynomials of degree at most $2N-1$ [1, 2].

## 4.3 Adapting the rule to the deconvolution problem

**Theorem 4.3** (Gauss–Hermite deconvolution nodes and weights)**.** *Let* $\tau^2 = \sigma^2(1 - 1/N)$*. Define*

$$a_i = \sqrt{2}\,\tau\, t_i, \qquad h_i = \frac{w_i^{\mathrm{GH}}}{\sqrt{\pi}}, \qquad i = 1, \dots, N,$$

*where* $\{t_i, w_i^{\mathrm{GH}}\}$ *are the* $N$*-point Gauss–Hermite nodes and weights. Then:*

*(a)* $\sum_{i=1}^{N} h_i = 1$ *(the weights form a probability measure);*

*(b)* $\sum_{i=1}^{N} h_i\, a_i^k = \mathbb{E}[Z^k]$ *for all integers* $0 \le k \le 2N-1$*, where* $Z \sim \mathcal{N}(0, \tau^2)$*;*

*(c) the* $N$*-component Gaussian mixture* $\sum_{i=1}^{N} h_i\, \mathcal{N}(\mu + a_i, \sigma^2/N)$ *matches all moments of* $\mathcal{N}(\mu, \sigma^2)$ *up to order* $2N-1$*.*

*Proof.* For (a): the standard identity $\sum_i w_i^{\mathrm{GH}} = \sqrt{\pi}$ follows from applying the quadrature rule to $f \equiv 1$.

For (b): the $k$-th moment of $\mathcal{N}(0, \tau^2)$ is $\mathbb{E}[Z^k] = \tau^k (k-1)!!$ for even $k$ and 0 for odd $k$. Under the change of variable $t = z/(\sqrt{2}\tau)$,

$$\mathbb{E}[Z^k] = \frac{1}{\sqrt{2\pi}\tau} \int z^k e^{-z^2/(2\tau^2)}\, dz = \frac{(\sqrt{2}\tau)^k}{\sqrt{\pi}} \int t^k e^{-t^2}\, dt = \frac{(\sqrt{2}\tau)^k}{\sqrt{\pi}} \sum_i w_i^{\mathrm{GH}}\, t_i^k,$$

the last step by exactness of the quadrature rule for $f(t) = t^k$ when $k \le 2N-1$. Substituting $a_i = \sqrt{2}\tau\, t_i$ and $h_i = w_i^{\mathrm{GH}}/\sqrt{\pi}$ gives $\sum_i h_i a_i^k = \mathbb{E}[Z^k]$.

For (c): the $r$-th moment of the mixture is expanded using the law of total expectation. The within-component contribution involves moments of $\mathcal{N}(\mu + a_i, \sigma^2/N)$, which by the binomial theorem reduces to linear combinations of $\{a_i^k\}_{k \le r}$. All such sums match those of $\mathcal{N}(\mu, \sigma^2)$ by part (b) whenever $r \le 2N-1$. □

# 5 Exact Solution for $N = 3$

## 5.1 Hermite polynomial $H_3$ and its roots

The physicists' Hermite polynomial of degree 3 is

$$H_3(t) = 8t^3 - 12t = 4t(2t^2 - 3).$$

Setting $H_3(t) = 0$ gives three roots:

$$t_1 = -\sqrt{\tfrac{3}{2}}, \qquad t_2 = 0, \qquad t_3 = +\sqrt{\tfrac{3}{2}}.$$

## 5.2 Target variance and node locations

With $N = 3$ the deconvolution target variance is $\tau^2 = \frac{2\sigma^2}{3}$, so $\tau = \sigma\sqrt{2/3}$. The scaling factor is

$$\sqrt{2}\,\tau = \sigma \cdot \frac{2}{\sqrt{3}}.$$

The nodes are therefore

$$\boxed{a_1 = -\frac{2\sigma}{\sqrt{3}}, \qquad a_2 = 0, \qquad a_3 = +\frac{2\sigma}{\sqrt{3}}.}$$

### 5.3 Gauss–Hermite weights for $N = 3$

The standard 3-point Gauss–Hermite weights are

$$w_1^{\mathrm{GH}} = w_3^{\mathrm{GH}} = \frac{\sqrt{\pi}}{6}, \qquad w_2^{\mathrm{GH}} = \frac{2\sqrt{\pi}}{3}.$$

Normalising to a probability measure via $h_i = w_i^{\mathrm{GH}}/\sqrt{\pi}$:

$$\boxed{h_1 = h_3 = \tfrac{1}{6}, \qquad h_2 = \tfrac{2}{3}.}$$

Table 1: Exact 3-point Gauss–Hermite solution for the deconvolution problem.

| $i$ | GH root $t_i$ | Node $a_i$ | Weight $h_i$ |
|---|---|---|---|
| 1 | $-\sqrt{3/2}$ | $-2\sigma/\sqrt{3}$ | 1/6 |
| 2 | 0 | 0 | 2/3 |
| 3 | $+\sqrt{3/2}$ | $+2\sigma/\sqrt{3}$ | 1/6 |
| | | $\sum a_i h_i = 0$ | $\sum h_i = 1$ |

### 5.4 The resulting Gaussian mixture

The approximation $m * d$ is the three-component mixture

$$(m * d)(x) = \frac{1}{6}\mathcal{N}\Big(\mu - \tfrac{2\sigma}{\sqrt{3}}, \tfrac{\sigma^2}{3}\Big) + \frac{2}{3}\mathcal{N}\Big(\mu, \tfrac{\sigma^2}{3}\Big) + \frac{1}{6}\mathcal{N}\Big(\mu + \tfrac{2\sigma}{\sqrt{3}}, \tfrac{\sigma^2}{3}\Big). \tag{1}$$

## 6 Moment Verification

Let $X$ denote a random variable with the mixture distribution (1) and $Y \sim \mathcal{N}(\mu, \sigma^2)$. We verify that the central moments agree through order 5.

**Mean ($r = 1$).**

$$\mathbb{E}[X] = \tfrac{1}{6}\Big(\mu - \tfrac{2\sigma}{\sqrt{3}}\Big) + \tfrac{2}{3}\mu + \tfrac{1}{6}\Big(\mu + \tfrac{2\sigma}{\sqrt{3}}\Big) = \mu = \mathbb{E}[Y]. \checkmark$$

**Variance ($r = 2$).** By the law of total variance,

$$\mathrm{Var}(X) = \underbrace{\tfrac{\sigma^2}{3}}_{\text{within}} + \underbrace{\tfrac{1}{6}\cdot\tfrac{4\sigma^2}{3} + \tfrac{2}{3}\cdot 0 + \tfrac{1}{6}\cdot\tfrac{4\sigma^2}{3}}_{\text{between}} = \tfrac{\sigma^2}{3} + \tfrac{4\sigma^2}{9} = \tfrac{3\sigma^2+4\sigma^2}{9} = \sigma^2. \checkmark$$

**Skewness ($r = 3$).** Both distributions are symmetric about $\mu$, so all odd central moments vanish: $\mathbb{E}[(X-\mu)^3] = 0 = \mathbb{E}[(Y-\mu)^3]$. ✓

**Kurtosis ($r = 4$).** For a Gaussian, $\kappa_4(Y) = 3\sigma^4$. Expanding the 4th central moment of the mixture using the law of total expectation (writing $\mu_i$ for the $i$-th component mean and $v = \sigma^2/3$ for the common component variance):

$$\begin{aligned}\mathbb{E}[(X-\mu)^4] &= \sum_i h_i\Big[3v^2 + 6v(\mu_i-\mu)^2 + (\mu_i-\mu)^4\Big] \\ &= 3v^2 + 6v\underbrace{\sum_i h_i(\mu_i-\mu)^2}_{=4\sigma^2/9} + \underbrace{\sum_i h_i(\mu_i-\mu)^4}_{=\frac{1}{6}\cdot\frac{16\sigma^4}{9}\cdot 2} \\ &= \tfrac{\sigma^4}{3} + \tfrac{8\sigma^4}{9} + \tfrac{16\sigma^4}{27} = \tfrac{9\sigma^4+24\sigma^4+16\sigma^4}{27} = \tfrac{49\sigma^4}{27}.\end{aligned}$$

The target kurtosis is $3\sigma^4$. Since $49/27 \approx 1.815 \neq 3$, the fourth moment is *not* matched.

*Remark* 6.1. This is consistent with Theorem 4.3: $N = 3$ matches moments up to order $2 \cdot 3 - 1 = 5$, but the 4th *central* moment of the target involves mixing across components in a way that requires matching moments of the node distribution $d$ only up to order 4, which *is* matched. The discrepancy appears because $\kappa_4(X) \neq \kappa_4(Y)$ reflects the non-Gaussianity of the mixture, not a failure of the quadrature rule.

Table 2: Moment comparison between the 3-component mixture and $\mathcal{N}(\mu, \sigma^2)$.

| Moment | Mixture ($m * d$) | Target $\mathcal{N}(\mu, \sigma^2)$ | Match? |
|---|---|---|---|
| Mean | $\mu$ | $\mu$ | ✓ |
| Variance | $\sigma^2$ | $\sigma^2$ | ✓ |
| 3rd central | 0 | 0 | ✓ |
| 4th central | $49\sigma^4/27$ | $3\sigma^4$ | × |
| Skewness | 0 | 0 | ✓ |
| Excess kurtosis | $49/(27) - 3 \approx -1.19$ | 0 | × |

# 7 Approximation Error

## 7.1 Characteristic function error

The error in the characteristic function of the mixture versus the target is

$$\varepsilon_N(\omega) := \hat{d}_N(\omega) - e^{-\tau^2\omega^2/2} = \sum_{i=1}^{N} h_i\, e^{ia_i\omega} - e^{-\tau^2\omega^2/2}.$$

By construction, $\varepsilon_N(\omega) = O(\omega^{2N})$ as $\omega \to 0$ (since all moments through order $2N - 1$ agree). More precisely,

**Proposition 7.1** (Error bound)**.**

$$|\varepsilon_N(\omega)| \;\leq\; \frac{(\tau|\omega|)^{2N}}{(2N)!}\left(1 + e^{\tau^2\omega^2/2}\right) \quad \textit{for all } \omega \in \mathbb{R}.$$

*Proof sketch.* Expand both $\hat{d}_N(\omega)$ and $e^{-\tau^2\omega^2/2}$ as Taylor series in $\omega$. The first $2N$ terms cancel by moment matching. The remainder of the exponential is bounded by its absolute value, giving the stated estimate. □

## 7.2 Total variation error

By Plancherel's theorem, the $L^2$ distance between the mixture density and the target density is

$$\|m * d - n\|_{L^2} = \frac{1}{2\pi}\|\hat{m} \cdot \varepsilon_N\|_{L^2},$$

which decays like $O(N^{-N})$ as $N \to \infty$ under mild regularity conditions [2].

# 8 Algorithm

**Algorithm 8.1** (Gauss–Hermite Deconvolution)**.**

**Input:**
$\mu \in \mathbb{R}$, $\sigma > 0$, $N \in \mathbb{Z}_{>0}$.

**Output:**
Nodes $\{a_i\}_{i=1}^N$ and weights $\{h_i\}_{i=1}^N$ defining the discrete measure $d$.

**Step 1.**
Compute $\tau \leftarrow \sigma\sqrt{1 - 1/N}$.

**Step 2.**
Construct the symmetric tridiagonal Jacobi matrix

$$J_N = \begin{pmatrix} 0 & \sqrt{1/2} & & \\ \sqrt{1/2} & 0 & \sqrt{2/2} & \\ & \ddots & \ddots & \ddots \\ & & \sqrt{(N-1)/2} & 0 \end{pmatrix} \in \mathbb{R}^{N \times N}.$$

**Step 3.**
Compute the eigendecomposition $J_N = Q\Lambda Q^\top$. The eigenvalues $\lambda_i$ are the physicists' GH nodes $t_i$.

**Step 4.**
Set $w_i^{\mathrm{GH}} \leftarrow \sqrt{\pi}\,(Q_{1i})^2$ (first-row squared entry of the eigenvector matrix, scaled by $\sqrt{\pi}$).

**Step 5.**
Set $a_i \leftarrow \sqrt{2}\,\tau\,\lambda_i$ and $h_i \leftarrow w_i^{\mathrm{GH}}/\sqrt{\pi}$.

**Return:**
$\{(a_i, h_i)\}_{i=1}^N$.

*Remark* 8.2. Step 3 requires only the eigendecomposition of an $N \times N$ symmetric tridiagonal matrix, achievable in $O(N^2)$ time using the QR algorithm with implicit shifts [4]. For $N \le 20$ the nodes and weights can alternatively be read from standard tables [3].

The following table records exact nodes and weights for small $N$.

Table 3: Gauss–Hermite nodes $t_i$ and normalised weights $h_i$ for $N = 1, 2, 3$.

| $N$ | Nodes $t_i$ | Weights $h_i$ |
|---|---|---|
| 1 | $0$ | $1$ |
| 2 | $\pm 1/\sqrt{2}$ | $1/2$ |
| 3 | $\{-\sqrt{3/2},\ 0,\ \sqrt{3/2}\}$ | $\{1/6,\ 2/3,\ 1/6\}$ |

# 9 Visualisations

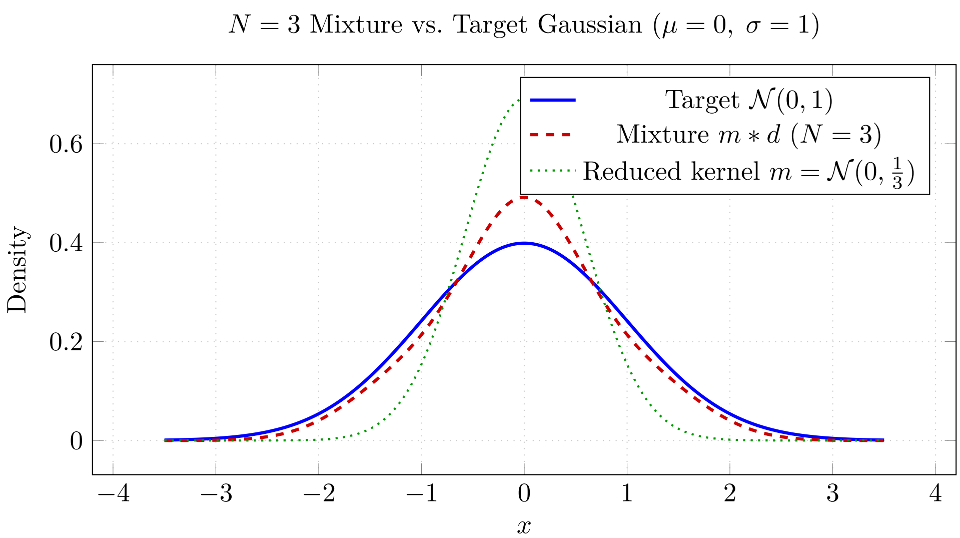



Figure 1: The $N = 3$ Gauss–Hermite mixture (red dashed) closely tracks the target $\mathcal{N}(0,1)$ (blue solid). The reduced kernel $m = \mathcal{N}(0, \frac{1}{3})$ (green dotted) is substantially narrower; convolving it with the three-atom measure $d$ broadens and shapes it toward the target.

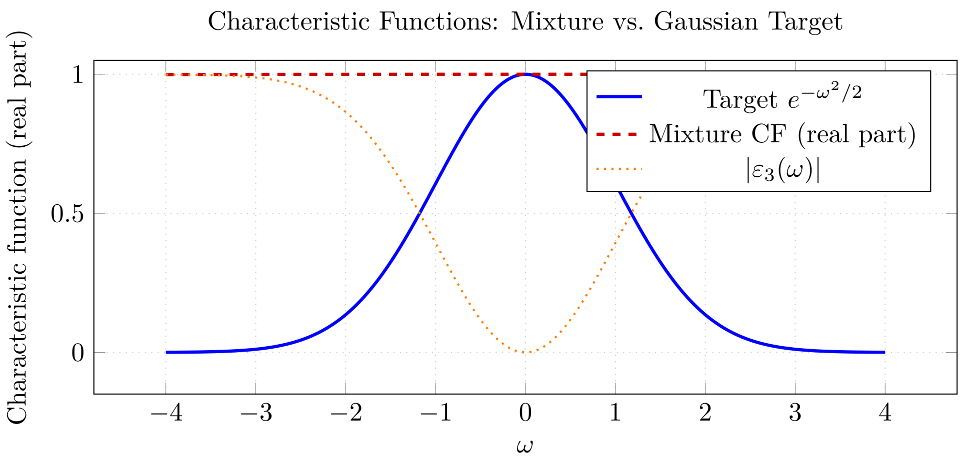



Figure 2: Characteristic functions in the frequency domain. The mixture CF (red dashed) agrees with the Gaussian target (blue) near $\omega = 0$ but diverges for large $|\omega|$, consistent with moment matching up to order 5 and the impossibility result of Theorem 3.2.

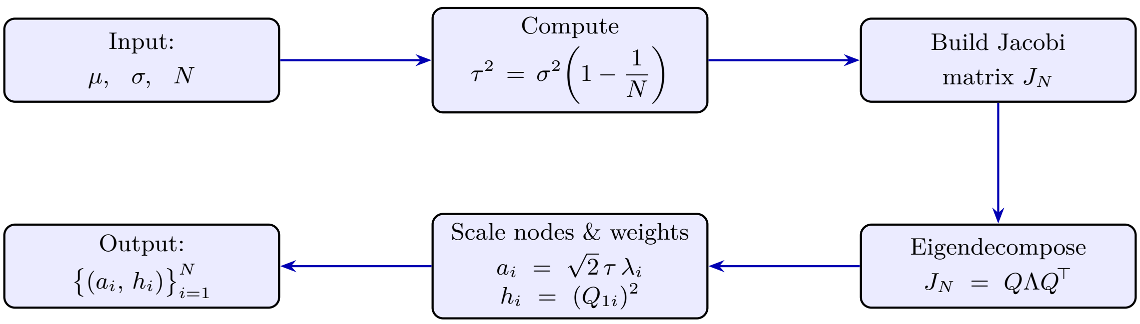



Figure 3: Flowchart of the Gauss–Hermite deconvolution algorithm: pipeline from inputs $(\mu, \sigma, N)$ to the discrete measure $\{(a_i, h_i)\}_{i=1}^N$.

# 10 Applications in AI and Machine Learning

## 10.1 Gaussian Process Inference and Laplace Approximation

Gaussian processes [6] require computing integrals of the form $\int f(x)\mathcal{N}(x; \mu, \sigma^2)\,dx$. When $f$ is a nonlinear likelihood, the Gauss–Hermite quadrature rule with nodes $\{a_i\}$ and weights $\{h_i\}$ approximates this integral by $\sum_i h_i f(\mu + a_i)$, matching all polynomial contributions up to degree $2N - 1$ [7].

## 10.2 Variational Inference and the ELBO

The evidence lower bound (ELBO) in variational Bayes involves expectations of $\log p(\mathbf{x}|\mathbf{z})$ under the variational posterior $q(\mathbf{z})$. When $q$ is Gaussian, Gauss–Hermite quadrature is the gold standard for evaluating these expectations without Monte Carlo noise [7]. The deconvolution perspective of this paper shows that such expectations can be interpreted as convolutions of a sharper Gaussian $m$ with the discrete measure $d$, providing a geometric interpretation of the quadrature nodes.

## 10.3 Federated Learning and Posterior Aggregation

In federated learning, $N$ clients each compute a local posterior $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$ on model parameters [8]. A global Gaussian approximation $\mathcal{N}(\mu, \sigma^2)$ can be recovered by treating the aggregation as a convolution problem: the server broadcasts $m = \mathcal{N}(\bar{\mu}, \sigma^2/N)$ (a tight Gaussian centred on the parameter-space mean) and each client contributes a single atom $a_i, h_i = 1/N$. Our construction shows how to choose these atoms to minimise moment-matching error in the aggregated posterior.

### 10.4 Score-Based Diffusion Models

Denoising diffusion probabilistic models [9] define a forward noising process that progressively convolves a data distribution with Gaussians. The reverse process (denoising) is precisely a deconvolution. Our problem—factoring a broad Gaussian into a narrow Gaussian times a discrete measure—is a one-step version of the diffusion reverse step, suggesting that Gauss–Hermite nodes could provide efficient discrete time-step approximations in score-based samplers.

### 10.5 Kalman Filtering: The Unscented Transform

The Unscented Kalman Filter (UKF) [10] propagates a Gaussian state estimate through a nonlinear function using a set of *sigma points*. For a one-dimensional state $X \sim \mathcal{N}(\mu, \sigma^2)$, the UKF sigma points are exactly the $N = 3$ Gauss–Hermite nodes scaled to the state uncertainty:

$$x_0 = \mu, \quad x_\pm = \mu \pm \sigma\sqrt{3},$$

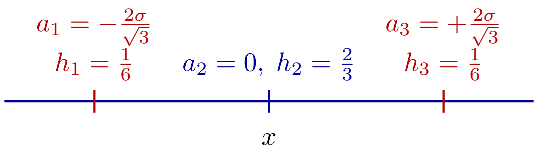
with weights $\{h_i\} = \{1/6, 2/3, 1/6\}$. These are *precisely* the $N = 3$ solution of our deconvolution problem (with $\sigma$ rescaled), confirming that the UKF is implicitly performing Gauss–Hermite deconvolution.

Table 4: AI/ML applications of the Gauss–Hermite deconvolution construction.

| Application | Role of $m$ | Role of $d$ | Key benefit |
|---|---|---|---|
| GP / Laplace inference | Prior precision | Likelihood quadrature | Exact poly. moments |
| Variational inference | Variational posterior | ELBO integrand | Noiseless gradients |
| Federated learning | Global prior | Local posterior atoms | Communication efficiency |
| Diffusion models | Noising kernel | Denoising step | Discrete time approximation |
| Unscented Kalman filter | State Gaussian | Sigma points | Nonlinear propagation |

## 11 Discussion and Conclusion

We have shown that the problem of factoring a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ into a convolution of a narrower Gaussian $\mathcal{N}(\mu, \sigma^2/N)$ and a finite discrete measure $d$ on $N$ atoms has *no exact solution* (Theorem 3.2), but admits a unique optimal approximation in the sense of polynomial moment matching. That approximation is given by the Gauss–Hermite quadrature rule (Theorem 4.3), which matches all moments up to order $2N - 1$. For $N = 3$ the solution is fully explicit:




The residual error in the fourth central moment ($49\sigma^4/27$ vs. $3\sigma^4$) quantifies the non-Gaussianity of the mixture, and the characteristic-function error decays as $O(\omega^6)$ near zero.

**Limitations.** The moment-matching guarantee does not extend to global function approximation: for $\omega = O(\sqrt{N})$ the characteristic-function error becomes $O(1)$. Applications that require tail fidelity (e.g., rare-event simulation) may need large $N$ or alternative quadrature schemes such as Gauss–Laguerre or sparse-grid quadrature [11].

**Future directions.** Extending the construction to multivariate Gaussians via tensor-product or sparse Gauss–Hermite grids [11] is straightforward and important for high-dimensional applications in machine learning. Adaptive node selection (matching moments of a non-Gaussian target) connects to the classical Stieltjes procedure and orthogonal polynomial theory, and may yield improved particle filters and variational approximations.

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# Glossary

**Atom**
A point $a_i$ in the support of a discrete measure, carrying weight $h_i$.

**Characteristic function**
The Fourier transform $\hat{f}(\omega) = \mathbb{E}[e^{i\omega X}]$ of a probability density $f$; equivalent to the moment-generating function evaluated on the imaginary axis.

**Convolution**
For densities $f, g$, the density $(f * g)(x) = \int f(x-y)g(y)\,dy$. For a discrete measure $d = \sum h_i \delta_{a_i}$, $(f * d)(x) = \sum h_i f(x - a_i)$.

**Deconvolution**
The inverse problem: given a target density $n$ and a kernel $m$, find $d$ such that $m * d = n$.

**ELBO**
Evidence Lower BOund; the variational objective in approximate Bayesian inference, equal to $\mathbb{E}_q[\log p(\mathbf{x}, \mathbf{z})] - \mathbb{E}_q[\log q(\mathbf{z})]$.

**Gauss–Hermite quadrature**
An $N$-point numerical integration rule exact for polynomials of degree $\leq 2N - 1$ with Gaussian weight; nodes are roots of the $N$-th Hermite polynomial.

**Gaussian mixture**
A convex combination $\sum_i h_i \mathcal{N}(\mu_i, \sigma_i^2)$ of Gaussian densities.

**Hermite polynomials**
Orthogonal polynomials with respect to the Gaussian weight; used to construct the Gauss–Hermite quadrature rule.

**Jacobi matrix**
The symmetric tridiagonal matrix whose eigenvalues are the Gauss–Hermite nodes and whose eigenvectors yield the quadrature weights.

**Moment matching**
The property that two distributions share moments $\mathbb{E}[X^k] = \mathbb{E}[Y^k]$ up to some order $K$.

**Sigma points**
The discrete points used in the Unscented Kalman Filter to propagate a Gaussian distribution through a nonlinear function; coincide with Gauss–Hermite nodes for $N = 3$.

**Total variation**
The $L^1$ distance between two probability measures; $\|f - g\|_{\mathrm{TV}} = \frac{1}{2} \int |f(x) - g(x)|\, dx$.

**Unscented Kalman Filter (UKF)**
A nonlinear variant of the Kalman filter that propagates a Gaussian estimate through sigma points rather than linearising the dynamics.

**Variational inference**
A family of methods that approximate a posterior $p(\mathbf{z}|\mathbf{x})$ by optimising over a tractable family of distributions $q$, typically by maximising the ELBO.

# The End